

# **A Multimodal Ontological Compendium for Cardiovascular Phenotyping: The Centrality of Cardiac Remodeling**

## **Preamble**

The modern practice of cardiology necessitates a departure from the serial, siloed interpretation of data. We must advance towards a unified, systems-based approach where clinical context, advanced imaging, and molecular biomarkers are synthesized into a coherent patient phenotype. Central to this endeavor is the concept of cardiac remodeling—the heart's integrated response to chronic hemodynamic stress or injury. As defined in a foundational consensus statement, cardiac remodeling is a process of genome expression resulting in molecular, cellular, and interstitial changes that manifest clinically as alterations in the size, shape, and function of the heart.<sup>1</sup> This process is not monolithic; it encompasses an initial, often adaptive phase aimed at maintaining cardiac function, which can transition into a maladaptive phase that drives the progression to heart failure (HF) and is associated with a poor prognosis, ventricular dysfunction, and malignant arrhythmias.<sup>1</sup> This compendium provides a structured, detailed analysis of the key variables used to map and interpret this critical process.

---

## **Part I: The Foundational Subject Node – Baseline Patient Attributes**

This section details the immutable or slowly changing patient characteristics that define a priori risk and serve as the primary triggers for the remodeling cascade.

### **Monograph 1.1: Demographic & Genetic Substrate**

### **1.1.1: Age: The Pathophysiology of Cardiac Senescence and Age-Related Remodeling**

Age is the single most significant non-modifiable risk factor for cardiovascular disease (CVD), reflecting the cumulative lifetime exposure to various risk factors as well as intrinsic processes of biological senescence.<sup>6</sup> The incidence of CVD increases dramatically with advancing age, rising from approximately 40% in individuals aged 40–59 years to around 75% in those aged 60–79, and reaching approximately 86% in those over the age of 80.<sup>6</sup>

This increased risk is paralleled by distinct structural and functional changes in the heart. With aging, both men and women exhibit a trend towards increased Left Ventricular (LV) mass and wall thickness.<sup>7</sup> However, the specific pattern of this age-related remodeling is fundamentally different between the sexes. In studies of Asian populations, for instance, women demonstrate a steeper rise in LV mass index with age, a change driven by increases in both wall thickness and chamber diameter. In contrast, the age-related increase in LV mass in men is primarily attributable to an increase in wall thickness alone.<sup>7</sup> This process of cardiac senescence is a key feature of the aging heart.<sup>8</sup> These structural alterations are accompanied by biochemical evidence of increasing stress. Serum levels of N-terminal pro-B-type natriuretic peptide (

NT-proBNP), a sensitive marker of myocardial wall stress and subclinical LV dysfunction, increase significantly with age, an association that is notably more pronounced in women.<sup>7</sup>

The observation that age-related remodeling follows sex-specific trajectories has profound clinical implications. The data indicate that the fundamental process of "cardiac aging" is not uniform. In women, aging appears to promote a remodeling pattern with a more prominent eccentric component (chamber dilation), which may be a key factor predisposing them to the higher rates of heart failure with preserved ejection fraction (HFpEF) observed in elderly females.<sup>7</sup> This is not simply an accumulation of disease but reflects an intrinsic, sex-differentiated biological aging process within the heart itself. Therefore, normative reference values for cardiac structure must be stratified by both age and sex to avoid the misinterpretation of

these physiological aging patterns as pathology.

### 1.1.2: Sex: A Comprehensive Analysis of Sex-Specific Differences in Cardiac Structure, Function, and Pathological Adaptation

Biological sex is a critical determinant of cardiovascular health, dictating distinct risk pathways, hormonal influences, and the normative reference values for cardiac structure and function. It is well-established that premenopausal women are generally protected against the adverse remodeling that occurs in response to pathological stimuli such as pressure or volume overload when compared to age-matched men.<sup>8</sup> This cardioprotection is largely attributed to the effects of estrogen and is significantly attenuated after menopause, which is associated with accelerated adverse LV remodeling, including increased myocardial fibrosis and diastolic dysfunction.<sup>6</sup>

The influence of sex hormones is a central mechanism. Estrogen and testosterone exert opposing effects on the remodeling heart.<sup>10</sup> Estrogen signaling, particularly through the estrogen receptor  $\beta$  (ER $\beta$ ), is protective against fibrosis and apoptosis. Conversely, androgens are associated with maladaptive hypertrophy and fibrosis, contributing to worse outcomes in certain conditions.<sup>9</sup> These hormonal differences lead to distinct pathological adaptations:

- **Pressure and Volume Overload:** In response to chronic pressure overload, women tend to develop a more concentric pattern of hypertrophy, characterized by wall thickening with smaller LV diameters. Men, in contrast, are more prone to maladaptive LV dilation.<sup>8</sup>
- **Myocardial Infarction (MI):** Men typically experience MI about a decade earlier than women. Following an MI, women have a higher risk of developing HF and exhibit higher mortality, even though their LVEF is often higher than in men. Structurally, men display larger post-MI LV chamber sizes and greater mass.<sup>8</sup>
- **Genetic Cardiomyopathies:** In hypertrophic cardiomyopathy (HCM), women often present at an older age but with more advanced disease, including more significant fibrosis and worse mortality.<sup>8</sup> In dilated cardiomyopathy (DCM), particularly with certain mutations like Titin, women often show less adverse remodeling and respond better to guideline-directed medical therapies.<sup>8</sup>

This evidence reveals that the "cardioprotective" female phenotype can be a double-edged sword. While premenopausal hormonal status confers a clear

advantage in common pathologies like hypertension, it may mask the progression of genetically driven diseases like HCM. In such cases, women present later, but their underlying disease is more severe and potentially less amenable to treatment.<sup>8</sup> Furthermore, because diagnostic thresholds for LV hypertrophy are often absolute or based on male-derived data, a woman's heart may need to undergo a greater relative degree of remodeling before it is classified as pathological, leading to diagnostic delays.<sup>9</sup> This suggests that the "protection" afforded by female sex hormones may, in some contexts, simply delay diagnosis, allowing the pathology to progress silently to a more advanced stage. This necessitates a higher index of suspicion and the development of sex-specific diagnostic criteria, particularly for genetic cardiomyopathies.

## **Monograph 1.2: Comorbid & Behavioral Substrate**

### **1.2.1: Diabetes Mellitus: Mechanisms of Diabetic Cardiomyopathy, Myocardial Fibrosis, and Microvascular Dysfunction**

Diabetes mellitus (DM) is a potent systemic risk factor that directly initiates and perpetuates adverse cardiac remodeling. This can lead to a specific clinical entity known as diabetic cardiomyopathy (DCM), which is defined by LV dysfunction that occurs in the absence of significant coronary artery disease, hypertension, or valvular disease.<sup>11</sup>

The pathophysiology of DCM is multifactorial, driven by a combination of metabolic and structural insults:

- **Myocardial Fibrosis and Stiffness:** DM is a powerful driver of myocardial fibrosis, both interstitial and perivascular, which increases myocardial stiffness and is a primary cause of the diastolic dysfunction characteristic of DCM.<sup>11</sup>
- **Metabolic Derangement and Lipotoxicity:** Systemic insulin resistance forces a metabolic shift in cardiomyocytes, which pivot from efficient glucose metabolism to fatty acid  $\beta$ -oxidation for energy. This shift is inefficient, increases metabolic pressure, generates excessive reactive oxygen species (ROS), and induces endoplasmic reticulum stress. The resulting accumulation of lipid intermediates

leads to lipotoxicity, promoting cardiomyocyte death (apoptosis).<sup>11</sup>

- **Advanced Glycation End-products (AGEs):** Chronic hyperglycemia promotes the non-enzymatic formation of AGEs. These products cross-link structural proteins like collagen within the myocardial interstitium, directly increasing the passive stiffness of the myocardium and impairing cardiac relaxation.<sup>11</sup>
- **Inflammation and Microvascular Dysfunction:** DM fosters a state of chronic, low-grade inflammation. It also promotes endothelial-to-mesenchymal transition (EndMT), a process where endothelial cells transform into fibroblasts, providing an additional source of collagen-producing cells that contribute to fibrosis.<sup>11</sup> In patients with coexisting aortic stenosis, DM is associated with more severe diffuse myocardial fibrosis and worse microvascular function, highlighting this synergistic damage.<sup>16</sup>

These mechanisms illustrate that diabetic cardiomyopathy represents a state of concurrent "energy crisis" and "structural stiffening." The cardiomyocyte itself is metabolically stressed and undergoing apoptosis due to the inefficient and toxic reliance on fatty acid oxidation, creating an energy crisis that underlies subclinical systolic dysfunction. Simultaneously, the extracellular matrix scaffolding the myocyte is becoming progressively stiffer due to AGE-induced collagen cross-linking, causing the structural stiffening that drives diastolic dysfunction. This elegant dual-pathway model explains why DCM characteristically presents with severe diastolic dysfunction but also harbors significant, often underappreciated, intrinsic systolic impairment, even when LVEF appears normal.

### 1.2.2: Smoking: The Vasculotoxic and Pro-inflammatory Impact on Endothelial Function and Cardiac Remodeling

Cigarette smoking is a major modifiable risk factor that exposes the cardiovascular system to a complex mixture of potent vasculotoxins and pro-inflammatory agents. These substances drive endothelial damage, a critical early event in atherogenesis that also contributes to adverse cardiac remodeling.<sup>17</sup>

The mechanisms of smoking-induced vascular injury are comprehensive:

- **Endothelial Dysfunction via Reduced Nitric Oxide (NO) Bioavailability:** Smoking, both acutely and chronically, impairs endothelium-dependent vasodilation. This is primarily achieved by reducing the bioavailability of NO, a

critical vasodilator and anti-inflammatory signaling molecule. The high levels of oxidative stress generated by cigarette smoke, particularly from superoxide anions, lead to the rapid scavenging and inactivation of NO. Furthermore, smoking can "uncouple" endothelial nitric oxide synthase (eNOS), the enzyme responsible for NO production, causing it to produce more superoxide instead of NO.<sup>17</sup>

- **Inflammation and Oxidative Stress:** Smoking induces a systemic and local inflammatory state, evidenced by elevated levels of biomarkers like high-sensitivity C-reactive protein (hs-CRP). It also increases the expression of vascular cell adhesion molecules (VCAM-1) on the endothelial surface, which promotes the adherence of leukocytes and initiates an inflammatory cascade within the vessel wall.<sup>17</sup> The thousands of chemical components in cigarette smoke, including free radicals, inflict direct oxidative damage on endothelial cells.<sup>24</sup>
- **Prothrombotic State:** By damaging the endothelium, smoking shifts the hemostatic balance towards a procoagulant state, increasing platelet and macrophage adherence and heightening the risk of thrombosis.<sup>17</sup>
- **Direct Cardiac Effects:** Beyond its vascular effects, chronic exposure to cigarette smoke has been demonstrated in animal models to directly cause hypertension and mild cardiac hypertrophy, linking its toxins to adverse structural remodeling of the heart itself.<sup>17</sup>

These pathways create a self-perpetuating cycle of vascular injury. Smoking initiates the process by causing direct endothelial dysfunction through multiple hits (reduced NO, increased ROS). This "activated" endothelium then expresses adhesion molecules, recruiting inflammatory cells. These cells, in turn, release more pro-inflammatory cytokines and ROS, which further amplify the endothelial damage and perpetuate the cycle.<sup>21</sup> This vicious feedback loop explains why smoking is such a powerful and accelerating risk factor for atherosclerosis; it does not inflict a single injury but rather ignites a self-sustaining pathological fire in the vessel wall. The documented improvement and even reversal of endothelial dysfunction after smoking cessation demonstrates that removing the initial trigger is effective at breaking this destructive cycle.<sup>26</sup>

### Monograph 1.3: Anthropometric & Hemodynamic Context

### 1.3.1: Body Surface Area (BSA): The Indispensable Role of Indexing in Cardiac Assessment

Anthropometric data are not merely descriptive but serve as essential denominators for normalization. Body Surface Area (BSA) is the indispensable standard for indexing cardiac structural and functional data. This practice is critical for distinguishing between physiological adaptation (e.g., in an athlete) and pathological remodeling, as it provides a better indication of metabolic mass than body weight alone.<sup>27</sup>

The clinical applications of BSA are foundational to modern cardiovascular assessment:

- **Cardiac Index (CI):** The most well-known application is the calculation of the Cardiac Index ( $CI = \text{Cardiac Output} / \text{BSA}$ ). This normalizes the heart's pumping capacity to the patient's body size and metabolic demand, allowing for standardized comparisons of cardiac function across individuals of different sizes.<sup>27</sup>
- **Indexing Structural Measures:** International guidelines mandate the indexing of LV, left atrial (LA), and aortic root dimensions to BSA. This is crucial for the accurate diagnosis of chamber enlargement and ventricular hypertrophy.<sup>32</sup>
- **Right Heart Assessment:** A strong positive correlation exists between right heart dimensions and BSA. This evidence suggests that right ventricular and atrial measurements should also be indexed to BSA to avoid misclassification, such as over-diagnosing RV dilation in a large but healthy individual or under-diagnosing it in a smaller person.<sup>32</sup>
- **Standardizing Imaging Protocols:** BSA has been shown to be a superior parameter to body weight for calculating optimal contrast media dosage in cardiac CT angiography, particularly in obese patients where weight-based dosing may be inaccurate.<sup>29</sup>

It becomes clear that failure to properly index by BSA introduces systemic error that propagates through the entire cardiovascular assessment pipeline. The influence of BSA begins with standardizing image *acquisition* (e.g., contrast dosing), continues through the interpretation of structural *measurements* (e.g., chamber sizes), and concludes with the assessment of global *function* (e.g., Cardiac Index). An un-indexed measurement in a large person could trigger a cascade of unnecessary and costly further investigations, while the same absolute value in a small person could lead to a

missed diagnosis and delayed treatment. Thus, the correct application of BSA is a foundational element for ensuring accuracy, comparability, and reliability across the entire diagnostic workflow.

---

## **Part II: The Myocardial Substrate – The Echocardiographic Phenotype**

This section provides an in-depth analysis of the structural and functional consequences of remodeling as quantified by echocardiography, the primary imaging modality for this purpose.

### **Monograph 2.1: Left Ventricular Systolic Function: The Engine's Efficacy**

#### **2.1.1: Left Ventricular Ejection Fraction (LVEF): The Historical Standard**

Left Ventricular Ejection Fraction (LVEF) has long been the most widely utilized parameter for assessing LV systolic function. It remains a cornerstone for therapeutic decision-making in heart failure, particularly for guiding the initiation of guideline-directed medical therapies in patients with reduced ejection fraction.<sup>2</sup>

However, its limitations as a true measure of myocardial health are well-recognized. LVEF is a relatively crude metric, representing the percentage change in LV chamber volume between diastole and systole, rather than a direct measure of intrinsic myocardial contractility.<sup>33</sup> Its calculation is often based on geometric assumptions that may not hold true in remodeled ventricles, and it is highly dependent on hemodynamic loading conditions; changes in preload or afterload can significantly alter

LVEF without any change in the heart muscle's intrinsic function. Critically, LVEF is often a late indicator in the continuum of cardiac disease, declining only after substantial myocardial damage has already occurred. Its prognostic value can be inconsistent, especially in patients with acute HF or in the increasingly prevalent

populations with HF with preserved (LVEF $\geq$ 50%) or mid-range (LVEF 40-49%) ejection fraction, where it often fails to predict mortality independently.<sup>33</sup>

### **2.1.2: Global Longitudinal Strain (GLS): A Superior Metric of Intrinsic Myocardial Contractility**

Global Longitudinal Strain (GLS), measured via speckle-tracking echocardiography, has emerged as a superior metric of intrinsic myocardial contractility. GLS provides a direct measure of myocardial fiber deformation, quantifying the percentage of shortening along the long axis of the ventricle. This is particularly sensitive because it primarily reflects the function of the subendocardial myocardial fibers, the layer most vulnerable to injury from ischemia, fibrosis, and pressure overload. A decline in GLS (i.e., a value becoming less negative, such as -16% instead of -20%) is frequently the earliest detectable sign of functional impairment due to maladaptive remodeling, often preceding any measurable drop in LVEF.<sup>33</sup>

The superiority of GLS over LVEF is most evident in its robust prognostic power. In large studies of patients with acute HF, GLS has demonstrated greater prognostic value than LVEF across the entire spectrum of heart failure—reduced, mid-range, and preserved.<sup>33</sup> While there is a moderate correlation between the two metrics, for any given

LVEF value, there is a wide distribution of GLS values, meaning that GLS can unmask significant contractile dysfunction even when LVEF is within the normal range.<sup>33</sup> In multivariable analyses,

GLS consistently emerges as an independent predictor of mortality, while LVEF often does not.<sup>33</sup> For instance, in acute HF, each 1% absolute improvement in

GLS is associated with a 5% decreased risk for mortality.<sup>33</sup> This prognostic strength extends to challenging populations; in patients with HFpEF, impaired

GLS is a robust predictor of adverse outcomes, and in patients with HF with *improved* EF (HFimpEF), a persistently abnormal GLS continues to predict future adverse events and functional decline, a role LVEF cannot fulfill.<sup>35</sup>

The widespread finding of abnormal GLS in patients with HFpEF fundamentally challenges the traditional understanding of this syndrome. The very definition of

HFpEF is predicated on a "preserved" LVEF, which has historically framed it as a disorder primarily of diastolic function. However, the evidence that the vast majority of HFpEF patients have significantly reduced (abnormal) GLS reveals a different story.<sup>33</sup> Because

GLS is a direct measure of longitudinal myocardial fiber shortening—that is, intrinsic contractility—its impairment signifies a failure of systolic function at the muscular level. The LVEF remains in the normal range due to compensatory mechanisms, such as preserved or enhanced radial contraction, or simply due to the geometric insensitivity of the LVEF calculation in a thick, non-compliant ventricle. This reframes HFpEF not as a pure diastolic problem, but as a syndrome of global myocardial failure with both systolic and diastolic components. This understanding is critical for the future development of effective therapies for this challenging condition.

Table 1: Prognostic Comparison of Key Systolic Function Metrics

| Feature                   | Left Ventricular Ejection Fraction (LVEF)         | Global Longitudinal Strain (GLS)                      |
|---------------------------|---------------------------------------------------|-------------------------------------------------------|
| What it Measures          | Volumetric change of the LV chamber (geometry)    | Myocardial fiber deformation (mechanics)              |
| Primary Determinant       | Endocardial border movement                       | Subendocardial fiber shortening                       |
| Load Dependency           | Highly dependent on preload and afterload         | Less load-dependent, more intrinsic                   |
| Sensitivity to Injury     | Late indicator; declines after significant damage | Early indicator; declines with subclinical injury     |
| Prognostic Value in HFrEF | Established, but can be inconsistent              | Superior predictor of all-cause mortality             |
| Prognostic Value in HFpEF | Poor; not independently predictive of mortality   | Robust; independent predictor of mortality/events     |
| Clinical Utility          | Cornerstone for guideline-based therapy           | Superior risk stratification across all HF phenotypes |

|  |            |  |
|--|------------|--|
|  | initiation |  |
|--|------------|--|

### Monograph 2.2: Left Ventricular Diastolic Function: The Engine's Ability to Fill

The assessment of left ventricular (LV) diastolic function is a cornerstone of modern echocardiography, providing critical insight into myocardial relaxation, compliance, and filling pressures. This assessment is central to understanding the transition from adaptive to maladaptive cardiac remodeling.

#### 2.2.1: The E/é Ratio: A Robust Non-Invasive Surrogate for LV Filling Pressures

The ratio of early mitral inflow velocity (E) to early diastolic mitral annular tissue velocity (é), known as the E/é ratio, is a cornerstone of modern diastolic function assessment. It serves as a robust and reliable non-invasive surrogate for estimating LV filling pressures. An elevated E/é ratio, particularly a value greater than 14, is highly specific for the presence of elevated LV end-diastolic pressure, a key driver of symptoms and adverse remodeling. Its prognostic significance has been validated in numerous clinical settings. In patients following a first ST-elevation MI (STEMI), the E/é ratio was the single most powerful predictor of developing acute, in-hospital HF.<sup>38</sup> Similarly, in patients with acute HFpEF, an E/é ratio exceeding 13.8 was found to be an independent predictor for rehospitalization, identifying a high-risk cohort.<sup>39</sup>

##### Monograph 2.2.2: The E, A, and E/A Waves: Foundational Mitral Inflow Patterns

The primary method for assessing LV diastolic function involves analyzing the transmitral inflow Doppler signal, which graphically displays the velocities of blood flow across the mitral valve during diastole. This signal is characterized by two distinct waves:

- The E Wave (Early Diastolic Velocity): This wave represents the rapid, passive filling of the left ventricle immediately after the mitral valve opens. It is driven by the pressure gradient between the left atrium and the relaxing left ventricle. A higher E wave velocity indicates a steeper pressure gradient, which can be influenced by LV relaxation, preload, and filling pressure.
- The E Wave Deceleration Time (DT): The DT is the duration from the peak of the E wave to its baseline. It reflects the rate of the decrease in pressure gradient between the left atrium and left ventricle. It is a measure of both LV relaxation and stiffness. A shortened DT (typically <150 ms) suggests a restrictive filling pattern with high left atrial pressure, while a prolonged DT (typically >220 ms) is a sign of impaired relaxation.
- The A Wave (Late Diastolic Velocity): This wave represents the active filling of the left ventricle due to atrial contraction. Its amplitude is directly related to the force of atrial systole. In patients with normal diastolic function, the E wave is typically dominant. However, with impaired LV relaxation, the E wave velocity decreases, and the heart becomes more dependent on atrial kick, leading to a prominent A wave.
- The E/A Ratio: The ratio of the peak E wave velocity to the peak A wave velocity is a fundamental marker of diastolic function.
  - Normal Function: E/A > 0.8

- Impaired Relaxation:  $E/A < 0.8$  (due to a decreased E wave and/or increased A wave)
- Pseudonormal Filling:  $E/A > 0.8$  (mimicking normal function, but driven by elevated filling pressures)
- Restrictive Filling:  $E/A > 2.0$  (signifying very high filling pressures and poor prognosis)

The E/A ratio is a key component of the American Society of Echocardiography (ASE) guidelines for grading diastolic dysfunction. However, it is a preload-dependent parameter, which is why it must be interpreted alongside other metrics, such as the more preload-independent E/e' ratio described in Monograph 2.2.1.

#### Semantic Annotations (SNOMED CT & UMLS)

For data standardization and interoperability, the following concepts are mapped to clinical terminologies:

- **Mitral Valve E-Wave Peak Velocity:**
  - UMLS CUI: [C0801086](#)
- **Mitral Valve A-Wave Peak Velocity:**
  - UMLS CUI: [C0801028](#)
- **Mitral Valve E to A Ratio:**
  - UMLS CUI: [C0801087](#)
- **Mitral Valve E-Wave Deceleration Time:**
  - SNOMED CT: [399354002](#)
  - UMLS CUI: [C1302337](#)

### 2.2.3: Left Atrial Volume Index (LAVi): The Morphological Barometer of Chronic Diastolic Burden

The Left Atrial Volume Index (LAVi) is the quintessential marker of *chronically* elevated LV filling pressures. An enlarged left atrium, typically defined as a  $LAVi > 34 \text{ mL/m}^2$ , is not a result of passive stretching. Instead, it represents the morphological consequence of active remodeling processes, including myocyte hypertrophy and interstitial fibrosis, that occur in the atrial wall in response to a sustained diastolic burden over time. It is, in effect, an integrated barometer of the pressure load the left atrium has been subjected to over months or years. An enlarged LA provides direct, structural evidence of long-standing diastolic dysfunction and, importantly, creates the arrhythmogenic substrate that promotes the development of atrial fibrillation. Its prognostic value is formidable; in post-STEMI patients, an elevated LAVi was the most powerful predictor of mortality during a six-month follow-up period.<sup>38</sup> Along with the E/e' ratio,

LAVi is a key parameter recommended by international guidelines for the comprehensive assessment of diastolic function and risk stratification.<sup>40</sup>

The combined assessment of E/e' and LAVi provides a powerful narrative of both the current hemodynamic state and the chronicity of the disease process. The E/e' ratio

can be thought of as a "weather report," offering a snapshot of the LV filling pressures at a single point in time. In contrast, LAVi functions as the "climate," reflecting the long-term structural consequences of those pressures. This distinction is clinically powerful. For instance, a patient presenting with an acutely elevated E/e ratio but a normal LAVi may be experiencing a new or transient hemodynamic issue. Conversely, a patient with a significantly increased LAVi has an established chronic problem, regardless of their E/e ratio at that moment. This is supported by clinical data showing that E/e is the best predictor of *acute, in-hospital* HF, whereas LAVi is the superior predictor of *long-term (6-month)* mortality.<sup>38</sup> Using these two parameters in concert allows clinicians to distinguish between acute, chronic, and acute-on-chronic diastolic pathophysiology with much greater precision.

## **Monograph 2.3: The Morphological Manifestation: Patterns of Cardiac Remodeling**

### **2.3.1: Concentric Hypertrophy: The Response to Pressure Overload**

Concentric hypertrophy is a distinct pattern of cardiac remodeling defined by an increase in LV mass index (LVMI) in the context of a normal LV end-diastolic volume index (LVEDVi) and an increased relative wall thickness ( $RWT > 0.42$ ).<sup>41</sup> This pattern is the classic adaptive response to a chronic pressure overload stimulus, such as systemic hypertension or aortic stenosis.<sup>43</sup> The underlying cellular mechanism involves the addition of new sarcomeres in

**parallel** to existing myofibrils. This process increases the cross-sectional area (width) of individual cardiomyocytes, resulting in a thickening of the ventricular walls without a significant increase in the overall chamber size.<sup>43</sup> The primary functional consequence of this geometric change is profound diastolic dysfunction, driven by the increased myocardial stiffness and impaired ventricular relaxation.<sup>44</sup>

### **2.3.2: Eccentric Hypertrophy: The Response to Volume Overload**

Eccentric hypertrophy is defined by an increase in LVMI that is accompanied by an increase in LVEDVi (i.e., chamber dilation) and a normal or reduced RWT( $\leq 0.42$ ).<sup>41</sup> This pattern is the archetypal response to a chronic volume overload stimulus, such as that seen in significant valvular regurgitation or in some forms of post-MI heart failure.<sup>43</sup> At the cellular level, the mechanism is the addition of new sarcomeres in

**series** along the length of the cardiomyocyte. This adaptation elongates the muscle fibers, leading to an increase in the overall size and volume of the ventricular chamber.<sup>43</sup> While initially compensatory, this geometric pattern provides the structural basis for progressive systolic dysfunction, as the ventricle becomes more spherical and thus hemodynamically less efficient.<sup>47</sup>

**Table 2: Differentiating Patterns of Pathological Left Ventricular Hypertrophy**

| Feature                          | Concentric Hypertrophy                                  | Eccentric Hypertrophy                          |
|----------------------------------|---------------------------------------------------------|------------------------------------------------|
| <b>Primary Stimulus</b>          | Pressure Overload (e.g., Hypertension, Aortic Stenosis) | Volume Overload (e.g., Valvular Regurgitation) |
| <b>Cellular Adaptation</b>       | Sarcomeres added in <b>parallel</b>                     | Sarcomeres added in <b>series</b>              |
| <b>Myocyte Morphology</b>        | Increased cell width (thicker)                          | Increased cell length (longer)                 |
| <b>LV Wall Thickness</b>         | Increased                                               | Normal or decreased                            |
| <b>LV Chamber Size</b>           | Normal or decreased                                     | Increased (dilated)                            |
| <b>Relative Wall Thickness</b>   | Increased ( $>0.42$ )                                   | Normal or decreased ( $\leq 0.42$ )            |
| <b>Primary Functional Impact</b> | Diastolic Dysfunction (impaired filling)                | Systolic Dysfunction (impaired pumping)        |

**2.3.3: Reverse Remodeling: A Critical Therapeutic Goal and Prognostic Indicator**

Reverse remodeling is a critical therapeutic objective, defined as the reduction in LV mass and volumes (e.g., a decrease in LVMI and LVEDVi) and an associated

improvement in function (e.g., an increase in LVEF) following the implementation of effective therapy.<sup>47</sup> This process is induced by therapies that interrupt the maladaptive neurohormonal and mechanical signaling pathways that drive adverse remodeling, such as guideline-directed medical therapies (GDMT) including ACE inhibitors, ARBs, beta-blockers, and sacubitril/valsartan, as well as device-based therapies like cardiac resynchronization therapy (CRT).<sup>47</sup>

The prognostic significance of achieving reverse remodeling is profound. It is one of the most powerful predictors of improved long-term survival, reduced hospitalizations, and enhanced functional capacity in patients with HF.<sup>48</sup> Studies have consistently shown that patients who demonstrate left ventricular reverse remodeling (LVRR) have dramatically lower mortality rates compared to those who do not.<sup>48</sup> The failure to achieve LVRR despite therapy is a marker of poor prognosis and should prompt a reassessment and potential intensification of the treatment regimen.<sup>48</sup>

Therefore, reverse remodeling should be viewed as the ultimate surrogate endpoint for therapeutic efficacy in heart failure. Its presence is more than just a positive finding on a follow-up echocardiogram; it is direct, measurable proof that a given therapy has successfully interrupted the underlying disease process and favorably altered its natural history. The structural and functional improvements seen in LVRR are the tangible result of mitigating the pathological stimuli driving heart failure progression. For this reason, the serial assessment for LVRR on follow-up imaging should be a standard component of chronic HF management, serving as a key performance indicator for the chosen therapeutic strategy and guiding future clinical decisions.

---

## **Part III: The Systemic Milieu – The Biochemical Signature of Remodeling**

This layer provides a molecular-level snapshot of the processes driving and resulting from cardiac remodeling, offering insights that complement imaging and clinical data.

### **Monograph 3.1: Myocardial Injury & Neurohormonal Stress**

### **3.1.1: High-Sensitivity Troponins (hs-cTn): Markers of Chronic, Low-Grade Myocyte Injury**

The advent of high-sensitivity cardiac troponin (hs-cTn) assays has fundamentally changed the interpretation of this biomarker. While a large, acute rise in troponin remains the cornerstone for diagnosing acute myocardial infarction, the ability to detect minute, chronic elevations has revealed a new application: quantifying ongoing, low-grade myocyte injury. These chronic low-level elevations of hs-cTnI or hs-cTnT are a direct biochemical correlate of the cellular stress, apoptosis, and increased membrane permeability inherent in the process of maladaptive remodeling.<sup>51</sup> This release is not necessarily from large-scale myocyte necrosis but can reflect a spectrum of sublethal injury, including the release of proteolytic degradation products or leakage from stressed but viable cells.<sup>51</sup> It is the biochemical signature of myocardial "wear and tear."

This conceptual shift transforms troponin from a binary "infarction" marker into an analog "myocardial stress" gauge. Older, less sensitive assays had a high detection threshold, meaning a positive result almost always signified a major necrotic event. In contrast, hs-cTn assays can detect troponin in a large proportion of the apparently healthy population.<sup>53</sup> These low-level, chronic elevations are not "false positives"; they are strongly and independently predictive of future adverse cardiovascular events, including the development of HF, coronary heart disease, and increased mortality.<sup>53</sup> In patients with established HF, chronically elevated hs-cTn levels are also powerful predictors of poor outcomes.<sup>51</sup> This means that troponin should be viewed as a continuous variable where any detectable level carries some degree of risk, and that risk increases as the level rises. It provides a quantitative measure of the intensity of the adverse remodeling process at the cellular level.

### **3.1.2: Natriuretic Peptides (BNP, NT-proBNP): Fluid Biomarkers of Myocardial Wall Stress**

B-type natriuretic peptide (BNP) and its N-terminal pro-hormone fragment

(NT-proBNP) are hormones released from the cardiac ventricles in direct response to myocardial wall tension and stretch—the primary mechanical stimulus for remodeling.<sup>54</sup> Consequently, their circulating levels directly correlate with the severity of the hemodynamic stress (both pressure and volume overload) imposed on the heart.<sup>54</sup>

Their clinical utility is extensive and well-established:

- **Diagnosis:** BNP and NT-proBNP are considered gold-standard biomarkers for the diagnosis of heart failure in patients presenting with dyspnea.<sup>55</sup>
- **Prognosis:** Elevated levels are among the most powerful predictors of hospitalization and mortality in patients with both acute and chronic HF.<sup>55</sup>
- **Therapy Guidance:** Because their levels respond dynamically to changes in wall stress, serial measurement allows for effective monitoring of therapeutic efficacy. A sustained reduction in NP levels following treatment is a sign of successful ventricular unloading and is associated with reverse remodeling and substantially improved clinical outcomes. Indeed, NP-guided therapy strategies have been shown to lead to more aggressive uptitration of GDMT and better clinical results compared to standard care.<sup>55</sup>

A useful clinical framework is to view natriuretic peptides as the heart's "neurohormonal cry for help." The stimulus for their release is excessive wall stress, and their biological actions—diuresis, natriuresis, and vasodilation—are all aimed at reducing that stress.<sup>54</sup> In the setting of heart failure, this compensatory system is overwhelmed, but the signal persists. The circulating level of NPs is therefore a direct, quantitative measure of how much the heart is straining against its load. A high level signifies that the heart is under severe duress. A falling level indicates that therapeutic interventions are working, the hemodynamic load is decreasing, and the heart's "cry for help" is diminishing. This framing underscores their value not as passive markers of disease, but as dynamic indicators of physiological stress that can be used to guide and titrate life-saving therapies.

## **Monograph 3.2: The Multi-Organ Axis – Drivers and Consequences**

### **3.2.1: Renal Function (Creatinine, eGFR): The Cardiorenal Syndrome**

The heart and the kidneys are intricately linked in a bidirectional relationship, where acute or chronic dysfunction in one organ can induce dysfunction in the other. This interplay is known as the Cardiorenal Syndrome (CRS).<sup>58</sup> CRS is classified into five types based on the primary organ of dysfunction and the acuity of the condition. Of particular relevance to cardiac remodeling are Type 2 CRS, where chronic HF leads to progressive chronic kidney disease (CKD) via mechanisms like low forward flow and renal venous congestion, and Type 4 CRS, where primary CKD promotes adverse cardiac remodeling (e.g., LVH, fibrosis) and HF through mechanisms like chronic volume overload, neurohormonal activation, and the effects of uremic toxins.<sup>58</sup>

Serum creatinine and the estimated glomerular filtration rate (eGFR) are the most common markers used to assess renal function, but they can be late and potentially misleading indicators in the setting of HF.<sup>58</sup> A common clinical conundrum is the rise in serum creatinine during the treatment of acute decompensated HF. This often leads to a phenomenon of "creatinine-phobia," where clinicians reduce or stop essential therapies like diuretics or RAAS inhibitors for fear of causing kidney injury. However, evidence suggests that a small to moderate rise in creatinine during aggressive diuresis may reflect effective decongestion and hemoconcentration rather than true tubular injury. When associated with clinical improvement, this change in creatinine is not linked to worse outcomes.<sup>58</sup> This supports a paradigm shift towards "permissive nephropathy," where a mild rise in creatinine is accepted, and even expected, as a sign of effective treatment, provided the patient is clinically decongesting. The focus must be on the overall clinical trajectory of the patient, not on an isolated creatinine value.

**Table 3: Classification and Pathophysiology of Cardiorenal Syndromes (CRS)**

| CRS Type      | Name              | Primary Dysfunction              | Secondary Dysfunction     | Pathophysiological Link / Clinical Example |
|---------------|-------------------|----------------------------------|---------------------------|--------------------------------------------|
| <b>Type 1</b> | Acute Cardiorenal | Acute Cardiac (e.g., Cardiogenic | Acute Kidney Injury (AKI) | Severe hypoperfusion, acute venous         |

|               |                        |                                                      |                                             |                                                                                                  |
|---------------|------------------------|------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------|
|               |                        | Shock)                                               |                                             | congestion.                                                                                      |
| <b>Type 2</b> | Chronic<br>Cardiorenal | Chronic Cardiac<br>(e.g., Chronic<br>HF)             | Chronic Kidney<br>Disease (CKD)             | Long-term low<br>output, chronic<br>venous<br>congestion,<br>neurohormonal<br>activation.        |
| <b>Type 3</b> | Acute<br>Renocardiac   | Acute Kidney<br>(e.g., AKI)                          | Acute Cardiac<br>Dysfunction                | Fluid overload,<br>electrolyte<br>shifts, uremic<br>toxins, acute<br>inflammation.               |
| <b>Type 4</b> | Chronic<br>Renocardiac | Chronic Kidney<br>(e.g., CKD)                        | Chronic Cardiac<br>Dysfunction<br>(LVH, HF) | Chronic<br>volume/pressure<br>overload, RAAS<br>activation,<br>anemia, uremic<br>cardiomyopathy. |
| <b>Type 5</b> | Secondary CRS          | Systemic<br>Condition (e.g.,<br>Sepsis,<br>Diabetes) | Both Cardiac &<br>Renal<br>Dysfunction      | Cytokine storm,<br>endothelial<br>dysfunction,<br>widespread<br>inflammation.                    |

### 3.2.2: Glycemic Control (HbA1c): Impact on Myocardial Stiffness

Chronic hyperglycemia, as quantified by glycated hemoglobin (HbA1c), is a potent, non-hemodynamic trigger for adverse cardiac remodeling. Poor glycemic control is significantly and independently associated with the presence and severity of LV diastolic dysfunction.<sup>62</sup> The principal mechanism involves the formation of Advanced Glycation End-products (AGEs), which non-enzymatically cross-link collagen and other proteins in the myocardial interstitium. This process directly increases the intrinsic stiffness of the myocardium, impairing its ability to relax and fill during diastole.<sup>12</sup> Clinical studies have demonstrated a strong, graded correlation between higher HbA1c levels and worsening parameters of diastolic function, such as a lower E/A ratio, a higher E/é ratio, and a larger

LAVi.<sup>63</sup>

### 3.2.3: Inflammation (hs-CRP): An Active Mediator of Remodeling

Systemic inflammation is not a passive bystander but an active participant in the pathophysiology of cardiac remodeling and heart failure. Pro-inflammatory cytokines are known to promote myocyte hypertrophy and interstitial fibrosis, directly contributing to adverse structural changes.<sup>66</sup> High-sensitivity C-reactive protein (hs-CRP) is a robust and widely available biomarker of systemic inflammation. While not specific to the cardiovascular system, elevated hs-CRP levels are strongly associated with the development, progression, and prognosis of HF.<sup>66</sup> An elevated hs-CRP level (e.g.,

≥2mg/L) is recognized as a "risk enhancer" in primary prevention guidelines and is a powerful predictor of HF hospitalization and mortality in diverse populations, including those with and without established CVD, and importantly, in patients with normal LVEF.<sup>68</sup>

## Part IV: Synthesis & Conclusion: The Unified, Actionable Phenotype

The pinnacle of clinical reasoning is the synthesis of these layers into a single, actionable patient narrative centered on the process of remodeling. This integrated framework, with remodeling as its central axis, transforms disparate data points into a coherent diagnosis, a nuanced prognosis, and a clear therapeutic roadmap. It allows clinicians to decipher the story the heart is telling, from the initial molecular insult to the final structural consequence, enabling a truly precise and proactive approach to cardiovascular medicine.

Consider a 68-year-old female with long-standing, sub-optimally controlled hypertension (**Subject Node**).

Her echocardiogram (**Phenotype**) demonstrates the classic pattern of concentric hypertrophy: a thickened ventricle with a high LVMI, a normal LVEDVi, and severe

diastolic dysfunction ( $E/e'$  of 20, LAVi of 48 mL/m<sup>2</sup>). Her LVEF is 65%, but her GLS is abnormal at -15%, unmasking an early contractile deficit that the LVEF fails to capture.

Her biomarkers (**Signature**) show a moderately elevated NT-proBNP (quantifying the diastolic stress from high filling pressures), a chronic low-level hs-Tropl elevation (signaling ongoing myocyte strain and injury from the pressure overload), and a normal Creatinine for now.

**The Unified Phenotype:** This is a precise diagnosis of **Hypertensive Heart Disease with Maladaptive Concentric Remodeling, resulting in severe diastolic dysfunction (HFpEF) and subclinical systolic dysfunction**. The therapeutic goal is not merely blood pressure control, but inducing **reverse remodeling**. Success is defined by reducing the LVMI, normalizing filling pressures (lowering  $E/e'$  and NT-proBNP), and halting the decline in GLS. This approach aims to improve symptoms and fundamentally alter the disease's natural history, which, if unchecked, will progress to overt heart failure and atrial fibrillation. The delta variables on her follow-up echocardiogram and biomarker panel will be the ultimate measure of therapeutic success.

### Sources des citations

1. Pathophysiological Basis of Cardiac Remodeling - Scholars Research Library, consulté le juin 23, 2025, <https://www.scholarsresearchlibrary.com/articles/pathophysiological-basis-of-cardiac-remodeling-105642.html>
2. Cardiac remodeling—concepts and clinical implications: a ... - JACC, consulté le juin 23, 2025, <https://www.jacc.org/doi/10.1016/S0735-1097%2899%2900630-0>
3. Cardiac remodeling-concepts and clinical implications: a consensus ..., consulté le juin 23, 2025, [https://www.researchgate.net/publication/12600317\\_Cardiac\\_remolding\\_concepts\\_and\\_clinical\\_implications\\_a\\_consensus\\_paper\\_from\\_an\\_International\\_Forum\\_on\\_Cardiac\\_Remodeling\\_Behalf\\_of\\_an\\_International\\_Forum\\_on\\_Cardiac\\_Remodeling](https://www.researchgate.net/publication/12600317_Cardiac_remolding_concepts_and_clinical_implications_a_consensus_paper_from_an_International_Forum_on_Cardiac_Remodeling_Behalf_of_an_International_Forum_on_Cardiac_Remodeling)
4. Cardiac Remodeling: Concepts, Clinical Impact, Pathophysiological Mechanisms and Pharmacologic Treatment - PMC, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4728597/>
5. Cardiac Remodeling: Concepts, Clinical Impact, Pathophysiological Mechanisms and Pharmacologic Treatment - PubMed, consulté le juin 23, 2025, <https://pubmed.ncbi.nlm.nih.gov/26647721/>
6. Cardiovascular Risks Associated with Gender and Aging - PMC - PubMed Central, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6616540/>
7. Age, Gender and Load-Related Influences on Left Ventricular Geometric

- Remodeling, Systolic Mid-Wall Function, and NT-ProBNP in Asymptomatic Asian Population | PLOS One, consulté le juin 23, 2025, <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0156467>
8. Hearts apart: sex differences in cardiac remodeling in health and disease - PMC, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11213513/>
  9. Hearts apart: sex differences in cardiac remodeling in health ... - JCI, consulté le juin 23, 2025, <https://www.jci.org/articles/view/180074>
  10. Hypertensive Cardiac Remodeling in Males and Females - American Heart Association Journals, consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/01.hyp.0000256279.49882.d8>
  11. Central role of cardiac fibroblasts in myocardial fibrosis of ... - Frontiers, consulté le juin 23, 2025, <https://www.frontiersin.org/journals/endocrinology/articles/10.3389/fendo.2023.1162754/full>
  12. The Role of Cardiac Fibrosis in Diabetic Cardiomyopathy: From Pathophysiology to Clinical Diagnostic Tools - MDPI, consulté le juin 23, 2025, <https://www.mdpi.com/1422-0067/24/10/8604>
  13. Diabetic Cardiomyopathy | Circulation Research - American Heart Association Journals, consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/circresaha.117.311586>
  14. Association Between Diffuse Myocardial Fibrosis by Cardiac Magnetic Resonance Contrast-Enhanced T1 Mapping and Subclinical Myocardial Dysfunction in Diabetic Patients | Circulation: Cardiovascular Imaging, consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/circimaging.111.965608?doi=10.1161/CIRCIMAGING.111.965608>
  15. First in a series on diabetes and the heart: The impact of diabetes on the heart, a broad perspective - European Society of Cardiology, consulté le juin 23, 2025, <https://www.escardio.org/Journals/E-Journal-of-Cardiology-Practice/Volume-14/First-in-a-series-on-diabetes-and-the-heart-the-impact-of-diabetes-on-the-heart-a-broad-perspective>
  16. Impact of diabetes on remodelling, microvascular function and exercise capacity in aortic stenosis | Open Heart, consulté le juin 23, 2025, <https://openheart.bmj.com/content/10/2/e002441>
  17. Smoking and Cardiovascular Disease: Mechanisms of Endothelial ..., consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/atvbaha.113.300156>
  18. Smoking and cardiovascular disease: mechanisms of endothelial dysfunction and early atherogenesis - PubMed, consulté le juin 23, 2025, <https://pubmed.ncbi.nlm.nih.gov/24554606/>
  19. Effects of tobacco cigarettes, e-cigarettes, and waterpipe smoking on ..., consulté le juin 23, 2025, <https://academic.oup.com/eurheartj/article/41/41/4057/5861975>
  20. Cigarette Smoking Is Related to Endothelial Dysfunction of Resistance, but Not Conduit Arteries in the General Population—Results From the Gutenberg Health Study - Frontiers, consulté le juin 23, 2025, <https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2021.674622/full>

21. Acute effects of cigarette smoke on Endothelial Nitric Oxide synthase, vascular cell adhesion molecule 1 and aortic intima media thickness. - F1000Research, consulté le juin 23, 2025, <https://f1000research.com/articles/10-396>
22. Smoking Is Associated With Epicardial Coronary Endothelial Dysfunction and Elevated White Blood Cell Count in Patients With Chest Pain and Early Coronary Artery Disease | Circulation, consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/circulationaha.106.641654>
23. Cigarette Smoking and Atherosclerotic Cardiovascular Disease - PMC - PubMed Central, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10918046/>
24. Chronic cigarette smoking causes hypertension, increased oxidative stress, impaired NO bioavailability, endothelial dysfunction, and cardiac remodeling in mice - American Journal of Physiology, consulté le juin 23, 2025, <https://journals.physiology.org/doi/10.1152/ajpheart.00868.2010>
25. Cigarette smoking, endothelial injury and cardiovascular disease - PMC, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC2517732/>
26. Effects of Smoking and Smoking Cessation on Endothelial Function: 1-Year Outcomes From a Randomized Clinical Trial | JACC, consulté le juin 23, 2025, <https://www.jacc.org/doi/10.1016/j.jacc.2010.03.002>
27. www.ncbi.nlm.nih.gov, consulté le juin 23, 2025, <https://www.ncbi.nlm.nih.gov/books/NBK559005/#:~:text=In%20cardiac%20care%2C%20BSA%20is,misdiagnosis%20or%20inappropriate%20therapeutic%20interventions.>
28. Body Surface Area - StatPearls - NCBI Bookshelf, consulté le juin 23, 2025, <https://www.ncbi.nlm.nih.gov/books/NBK559005/>
29. Contrast Enhancement in Cardiovascular MDCT: Effect of Body Weight, Height, Body Surface Area, Body Mass Index, and Obesity | AJR - American Journal of Roentgenology, consulté le juin 23, 2025, <https://ajronline.org/doi/10.2214/AJR.07.2765>
30. Cardiac index - Wikipedia, consulté le juin 23, 2025, [https://en.wikipedia.org/wiki/Cardiac\\_index](https://en.wikipedia.org/wiki/Cardiac_index)
31. Physiology, Cardiac Index - StatPearls - NCBI Bookshelf, consulté le juin 23, 2025, <https://www.ncbi.nlm.nih.gov/books/NBK539905/>
32. Redalyc.Should we Quantify right Heart Diameters in relation to body Surface Area?, consulté le juin 23, 2025, <https://www.redalyc.org/pdf/3053/305355054007.pdf>
33. Global Longitudinal Strain to Predict Mortality in Patients With Acute ..., consulté le juin 23, 2025, <https://www.jacc.org/doi/10.1016/j.jacc.2018.02.064>
34. Global Longitudinal Strain to Predict Mortality in Patients With Acute Heart Failure - PubMed, consulté le juin 23, 2025, <https://pubmed.ncbi.nlm.nih.gov/29724346/>
35. Prognostic Value of LV Global Longitudinal Strain by 2D and 3D ..., consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/CIRCIMAGING.124.016975>
36. Prognostic Value of Global Longitudinal Strain in Patients With Heart Failure With Improved Ejection Fraction | JACC, consulté le juin 23, 2025, <https://www.jacc.org/doi/abs/10.1016/j.jchf.2021.08.007>

37. Prognostic Value of Global Longitudinal Strain in Patients With Heart Failure With Improved Ejection Fraction | JACC, consulté le juin 23, 2025, <https://www.jacc.org/doi/10.1016/j.jchf.2021.08.007>
38. Left Atrial Volume Index (LAVI) as a Predictor of Outcome for First ..., consulté le juin 23, 2025, [https://mjcu.journals.ekb.eg/article\\_57514\\_2a187b921d99f38309c2b1bdf5a8c052.pdf](https://mjcu.journals.ekb.eg/article_57514_2a187b921d99f38309c2b1bdf5a8c052.pdf)
39. The E/e' Ratio—Role in Risk Stratification of Acute Heart Failure with Preserved Ejection Fraction - PubMed Central, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8070491/>
40. The impact of diastolic dysfunction as a predictor of cardiovascular events: A systematic review and meta-analysis, consulté le juin 23, 2025, <https://www.revportcardiol.org/index.php?p=revista&tipo=pdf-simple&pii=S2174204920300465>
41. Understanding LVH Part 1: Concentric, Eccentric and Concentric Remodeling - Cardioserv, consulté le juin 23, 2025, <https://www.cardioserv.net/understanding-lvh-part-1-concentric-eccentric-concentric-remodeling/>
42. Concentric vs. eccentric remodelling in heart failure with reduced ejection fraction: clinical characteristics, pathophysiology and response to treatment - PubMed Central, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7540540/>
43. Ventricular hypertrophy - Wikipedia, consulté le juin 23, 2025, [https://en.wikipedia.org/wiki/Ventricular\\_hypertrophy](https://en.wikipedia.org/wiki/Ventricular_hypertrophy)
44. Cardiac remodelling: concentric versus eccentric hypertrophy in ..., consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC2300466/>
45. Molecular pathways underlying cardiac remodeling during ..., consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3076218/>
46. Cardiac Hypertrophy: Molecular and Cellular Events - Revista Española de Cardiología, consulté le juin 23, 2025, <https://www.revespcardiol.org/en-cardiac-hypertrophy-molecular-cellular-events-articulo-13088668>
47. Sacubitril/valsartan reduces levels of procollagen types I and III and correlates with reverse cardiac remodeling | REC: CardioClinics, consulté le juin 23, 2025, <https://www.reccardioclinics.org/en-sacubitril-valsartan-reduces-levels-procollagen-types-articulo-S2605153220300789>
48. Brasil - Reverse Cardiac Remodeling: A Marker of Better ... - SciELO, consulté le juin 23, 2025, <https://www.scielo.br/j/abc/a/n3PyVRT53bh9zPFWd9z33HB/>
49. Predictors of Spontaneous Reverse Remodeling in Mild Heart Failure Patients With Left Ventricular Dysfunction | Circulation, consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/circheartfailure.113.000929>
50. Incidence and prognosis implications of long term left ventricular ..., consulté le juin 23, 2025, [https://www.researchgate.net/publication/284232346\\_Incidence\\_and\\_prognosis\\_implications\\_of\\_long\\_term\\_left\\_ventricular\\_reverse\\_remodeling\\_in\\_patients\\_with\\_](https://www.researchgate.net/publication/284232346_Incidence_and_prognosis_implications_of_long_term_left_ventricular_reverse_remodeling_in_patients_with_)

dilated\_cardiomyopathy

51. Circulating Cardiac Troponin I in Severe Congestive Heart Failure, consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/01.CIR.96.9.2953>
52. Type 2 MI and Myocardial Injury in the Era of High-sensitivity Troponin | ECR Journal, consulté le juin 23, 2025, [https://www.ecrjournal.com/articles/type-2-mi-and-myocardial-injury-era-high-sensitivity-troponin?language\\_content\\_entity=en](https://www.ecrjournal.com/articles/type-2-mi-and-myocardial-injury-era-high-sensitivity-troponin?language_content_entity=en)
53. High sensitivity troponins: A potential biomarkers of cardiovascular ..., consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9748104/>
54. Natriuretic Peptide B Type Test - StatPearls - NCBI Bookshelf, consulté le juin 23, 2025, <https://www.ncbi.nlm.nih.gov/books/NBK556136/>
55. NT-proBNP — A New Test for Diagnosis, Prognosis and ..., consulté le juin 23, 2025, [https://www.uscjournal.com/articles/nt-probnp-new-test-diagnosis-prognosis-and-management-congestive-heart-failure?language\\_content\\_entity=en](https://www.uscjournal.com/articles/nt-probnp-new-test-diagnosis-prognosis-and-management-congestive-heart-failure?language_content_entity=en)
56. BNP and NT-proBNP as Diagnostic Biomarkers for Cardiac Dysfunction in Both Clinical and Forensic Medicine - PMC, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6515513/>
57. Are Serial BNP Measurements Useful in Heart Failure Management? | Circulation, consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/circulationaha.112.120485>
58. Cardiorenal Syndrome in Heart Failure with Preserved Ejection ..., consulté le juin 23, 2025, <https://karger.com/crm/article/15/1/41/918715/Cardiorenal-Syndrome-in-Heart-Failure-with>
59. Cardiorenal Syndrome | JACC, consulté le juin 23, 2025, <https://www.jacc.org/doi/10.1016/j.jacc.2008.07.051>
60. Cardiorenal Syndrome and Heart Failure—Challenges and Opportunities - PMC - PubMed Central, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9257995/>
61. Cardiorenal Syndrome: Classification, Pathophysiology, Diagnosis, and Treatment Strategies: A Scientific Statement From the American Heart Association, consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/CIR.0000000000000664>
62. Impact of glycemic control on left ventricular diastolic dysfunction in ..., consulté le juin 23, 2025, <https://www.thepharmajournal.com/archives/2016/vol5issue4/PartB/14-2-9-759.pdf>
63. Effect of HbA1c on The Severity of Diastolic Dysfunction in Type Two Diabetic Patients Presenting to the Cardiology Department, - Pakistan Heart Journal, consulté le juin 23, 2025, <https://pakheartjournal.com/index.php/pk/article/download/2749/1543/14462>
64. Association between glycosylated hemoglobin levels, diabetes duration, and left ventricular diastolic dysfunction in patients with type 2 diabetes and preserved ejection fraction: a cross-sectional study, consulté le juin 23, 2025,

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC10761463/>
65. Elevated HbA1c Levels and Diastolic Discordance in Type 1 Diabetes: A Cardiac Echo Perspective, consulté le juin 23, 2025, <http://ijlbpr.com/uploadfiles/1vol13issue7pp1-5.20240701100432.pdf>
66. Role of Inflammation in Cardiac Remodeling After Acute ... - Frontiers, consulté le juin 23, 2025, <https://www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2022.927163/full>
67. Reappraisal of Inflammatory Biomarkers in Heart Failure - PMC, consulté le juin 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7017677/>
68. High-Sensitivity C-Reactive Protein Is Associated With Heart Failure ..., consulté le juin 23, 2025, <https://www.ahajournals.org/doi/10.1161/JAHA.123.032997>
69. High-sensitivity C-reactive Protein in Atherosclerotic Cardiovascular Disease: To Measure or Not to Measure? - US Cardiology Review (USC), consulté le juin 23, 2025, [https://www.uscjournal.com/articles/high-sensitivity-c-reactive-protein-atherosclerotic-cardiovascular-disease-measure-or-not?language\\_content\\_entity=en](https://www.uscjournal.com/articles/high-sensitivity-c-reactive-protein-atherosclerotic-cardiovascular-disease-measure-or-not?language_content_entity=en)